

Tech Demo Flight Plan

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ArchaeoSight

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CS Capstone Design

Technical Demo Grading Sheet (100 pts)

TEAM: ArchaeoSight

Overview: The main purpose of the “Technical Demos” is to very clearly communicate the extent to which the team has identified key challenges in the project, and has proven solutions to those challenges. Grading is based on how complete/accurate the list of challenges is, , and how convincingly and completely the given demos cover the given challenges.

This template is fleshed out by the team, approved by CS mentor, and brought to demo as a grading sheet.

Risky technical challenges

Based on our requirements acquisition work and current understanding of the problem and envisioned solution, the following are the key technical challenges that we will need to overcome in implementing our solution:

C1: ML model that can work in the app. While making the ML model may not be extremely challenging, making sure the pre-trained model is accurate and can work on the mobile device is crucial. This will allow for archaeologists to use the model in the field.

C2: Display of Various Data Types. Dr. Sharp works with different data formats, including pXRF output (.csv/.xlsx), 3D models (.glb/.gltf), and 2D images (.png/.jpg). Our app should be capable of displaying all of these formats to the user in a well-presented manner.

C3: Offline/Online ML Integration. Ensuring that our machine learning model functions reliably in both offline and online modes is a key challenge. The model must provide predictive insight in the field, without an internet connection. The complexity of the model increases with the need for responsiveness, accuracy, and minimal device performance impact.

C4: Importing & Validation of Data. ArchaeoSight must be able to receive data from different models of the pXRF (portable X-ray fluorescence spectrometer) such as the Sci-aps X-200 and Bruker Tracer models. This data is exported as CSV files from equipment through Bluetooth and USB connection. Some models of the pXRF preprocess their data before exportation making it easier to read and adapt to our system, other models that lack this feature may need additional processing before utilizing the collected data

Challenges covered by demos:

In this section, we outline the demonstrations we have prepared, and exactly which of the challenge(s) each one of them proves a solution to.

Demonstration 1: Data Display

Challenges addressed: C2: Display of Various Data Types

Flight Plan: Step by step overview of demo

1. Navigate to Data page of ArchaeoSight app

2. Select data to view
3. Repeat for each data type (pXRF output, 3D model, 2D image)

Evaluation:

- ✓ Proved we could display the data, no matter the format we accept.

Demonstration 2: GDBT ML Model

Challenges addressed: C3: Offline/Online ML Integration, C1: An ML model that can predict areas of archaeological/anthropological importance

Flight Plan: Step by step overview of demo

1. Launch ArchaeoSight App in Airplane Mode
2. Load pretrained TensorFlow Lite Model Locally
3. Input Offline Sample Data
4. Run Prediction Offline
5. Store Results Locally

Evaluation:

- ✓ Proved full offline functionality of ML inference and data capture.
- ✓ Confirmed uninterrupted user experience across network state transitions.

Demonstration 3: pXRF Data Integration

Challenges addressed: C4: Importing & Validation of data

Flight Plan: Step by step overview of demo

1. Launch ArchaeoSight Application
2. Connect pXRF tool via USB
3. Validate if data is preprocessed after exportation
4. Purify collected data if not preprocessed
5. View collected & processed data in app

Evaluation:

- ✓ Displayed data is formatted correctly on screen
- ✓ Collected data is adapted to be readable by ML model

Other challenges recognized by not addressed by demo:

If there were challenges you listed earlier that were *not* covered by a demo, list here. This will hopefully be a short list...but better to be clear about where you are. If you have items here, you could list (if applicable) any pending plans to reduce these risks.

- User authentication
- Data Syncing – We don't have “new” data to add to the cloud.